

VIIth Marseille International Cosmology Conference

The impact of He II reionisation on the H I Ly- α forest

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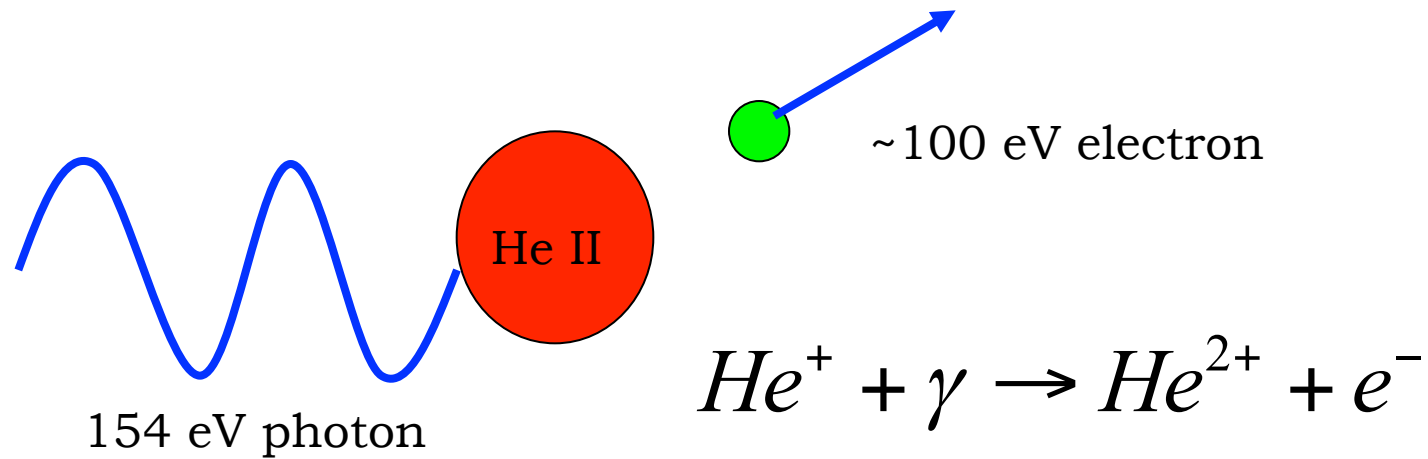
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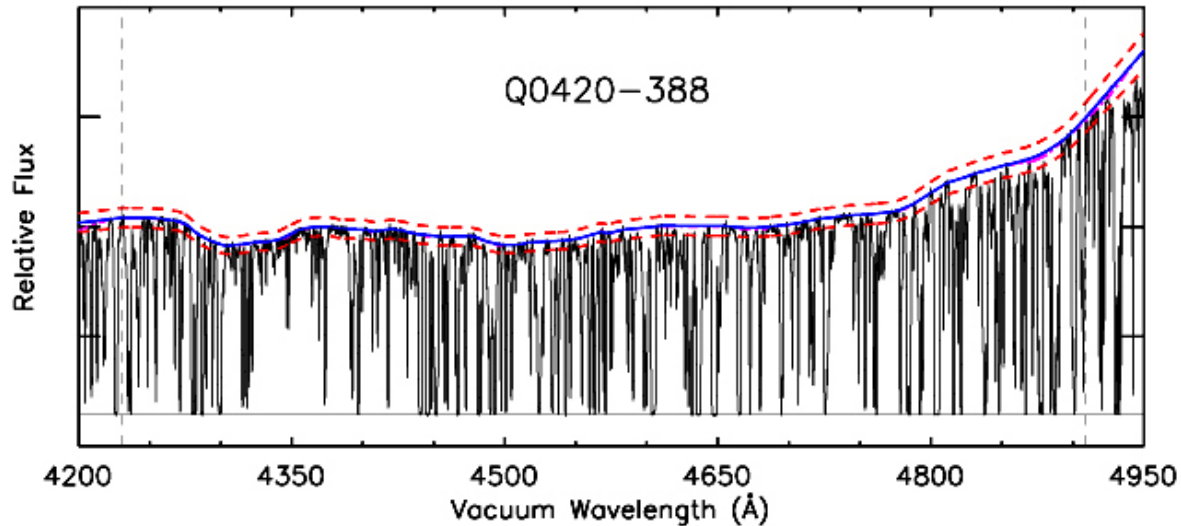
Photo-heating

Photons not only ionise – if they have $E > 54.4 \text{ eV}$ then they also heat the IGM.



Electrons share their energy with the baryons via Coulomb scattering.

The HI Ly- α forest at $z \sim 3$



Kim *et al.* (2007)

$$\tau = \tau_0 \frac{(1+z)^6 (\Omega_b h^2)^2}{T^{0.7} H(z) \Gamma(z)} \Delta^2$$

- Higher temperatures broaden the absorption lines and decrease the opacity.
- The HI forest is a very valuable *indirect* probe of He II reionisation through the IGM thermal state

Outline

- 1) The IGM temperature during He II reionisation (How much? How quickly?)
- 2) Implications for observations of the H I Ly- α forest opacity

How much heating?

RADIATIVE TRANSFER EFFECTS DURING PHOTOHEATING OF THE INTERGALACTIC MEDIUM

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Received 1999 March 8; accepted 1999 May 18; published 1999 June 22

ABSTRACT

The thermal history of the intergalactic medium (IGM) after reionization is to a large extent determined by photoheating. Here we demonstrate that calculations of the photoheating rate which neglect radiative transfer effects substantially underestimate the energy input during and after reionization. The neglect of radiative transfer effects results in temperatures of the IGM which are too low by a factor of 2 after He II reionization. We briefly discuss implications for the absorption properties of the IGM and the distribution of baryons in shallow potential wells.

Subject headings: cosmology: theory — galaxies: formation — quasars: absorption lines

Power law QSO spectrum $J_\nu \propto \nu^{-\alpha}$

Optically thin

$$\sigma_\nu \propto \nu^{-3} \Rightarrow \langle E \rangle = \frac{h\nu_{th}}{\alpha + 2}, \quad \Delta T \approx 4200K$$

Approximate optically thick limit

$$\sigma_\nu \propto \nu^0 \Rightarrow \langle E \rangle = \frac{h\nu_{th}}{\alpha - 1}, \quad \Delta T \approx 30000K$$

The mean free path is key

The amount of filtering depends on how clumpy the IGM is:

Mean free path: $\lambda_\nu \propto \nu^\beta$

$$\langle E \rangle = \frac{\int_{\nu_{th}}^{\infty} d\nu h(\nu - \nu_{th}) \epsilon_\nu \lambda_\nu \sigma_\nu}{\int_{\nu_{th}}^{\infty} d\nu \epsilon_\nu \lambda_\nu \sigma_\nu} \approx \frac{h \nu_{th}}{(\alpha - \beta + 2)}$$

$\beta=3$ for a uniform optically thick IGM, while optically thin has $\beta=0$ (frequency independent mean free path)

β will be intermediate between these two cases – hard photons are preferentially absorbed in dense regions, and the abundance of He II LLS is therefore crucial.

How quickly?

Filtering of ionising radiation not only **hardens** the spectrum – it also **weakens** the intensity.

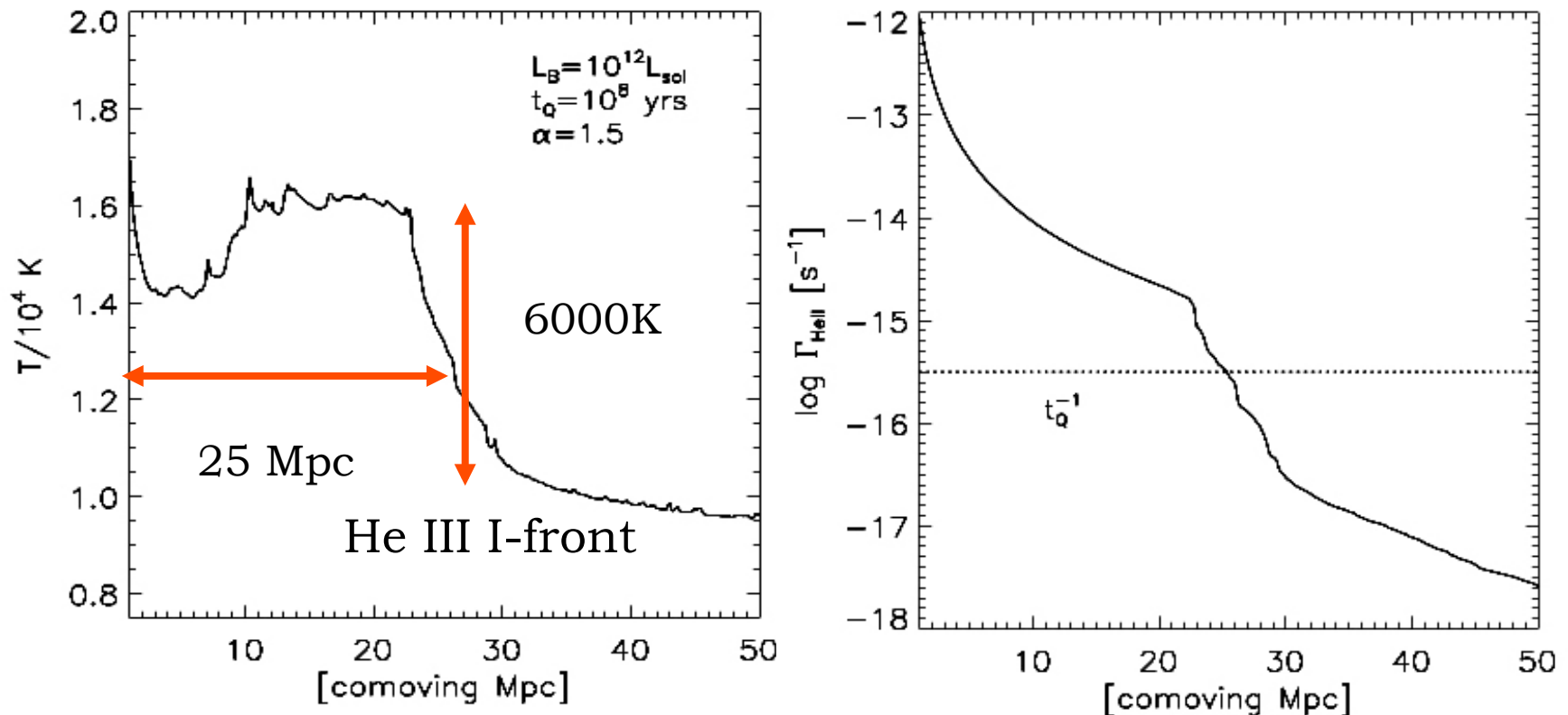
$$\frac{dT}{dt} \approx \frac{2\mu m_H}{3k_B\rho} \left| \frac{dn_{HeII}}{dt} \right| \langle E \rangle, \quad \left| \frac{dn_{HeII}}{dt} \right| \propto \frac{1}{t_{ion}}$$

Depends on radiation intensity

Depends on incident spectrum only

If $t_{ion} \gg t_Q$, no significant photo-heating will occur, even if $\langle E \rangle$ is very large. This limits the maximum temperature boost achievable in the IGM over a given timescale.

Line-of-sight radiative transfer



- Density distribution from high resolution Gadget-2 hydrodynamical simulation
- Multi-frequency line-of-sight radiative transfer applied to the density field in post processing.

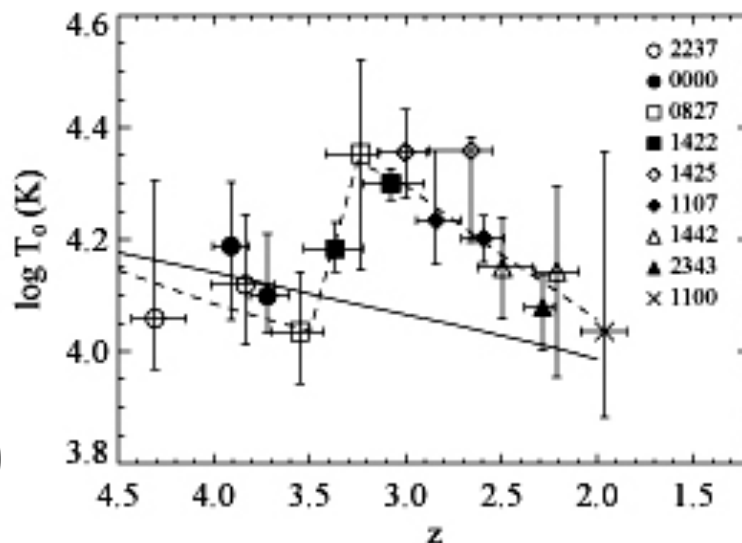
Bolton, Oh & Furlanetto (2009a)

Key points

- The *entire* IGM cannot be *rapidly* heated ($\Delta z \sim 0.1-0.2$), and not by $>10^4\text{K}$ unless the average quasar spectrum is much harder than $\langle\alpha\rangle=1.5$ – there are too few hard photons to achieve this quickly.
- Most heating in He III regions is significantly closer to the optically thin limit (although a weak, hard but persistent UVB may still heat isolated patches to much higher temperatures towards the end of reionisation).

Not necessarily in conflict with observations. The error bars are large and allow for more gradual photo-heating.

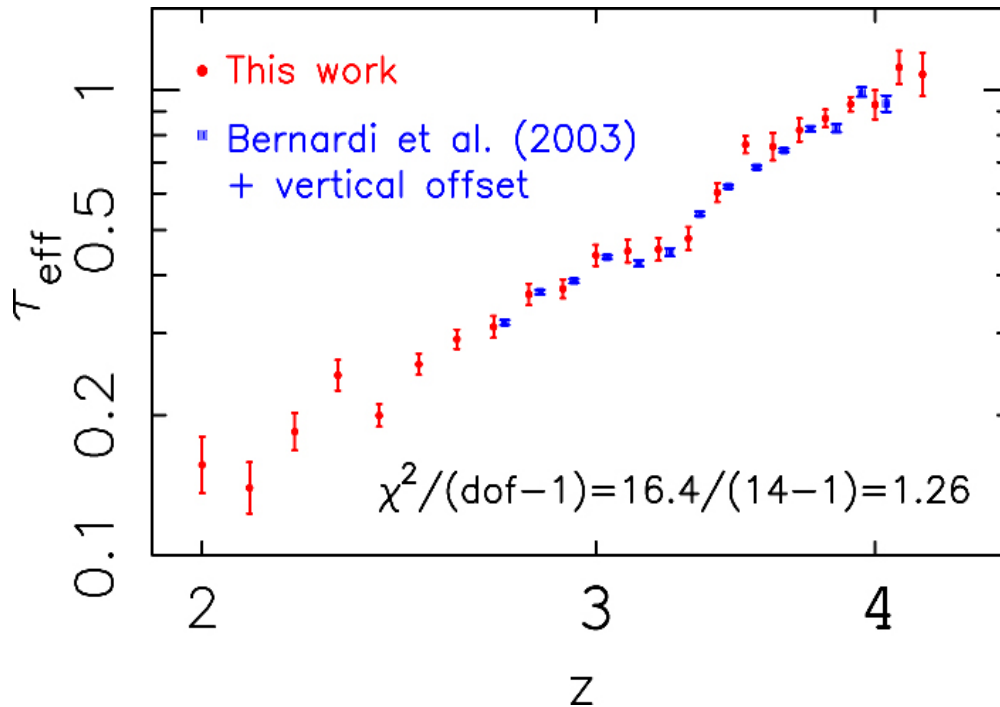
Schaye *et al.* (2000)



Outline

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The mysterious Ly- α opacity “dip”



$$\tau_{\text{eff}} = -\ln \langle F \rangle$$

Faucher-Giguere *et al.* (2008)

Feature is seen in 3 independent data sets

Bernardi *et al.* (2003): 1061 SDSS spectra

Faucher-Giguere *et al.* (2008): 86 HIRES/MIKE/ESI spectra

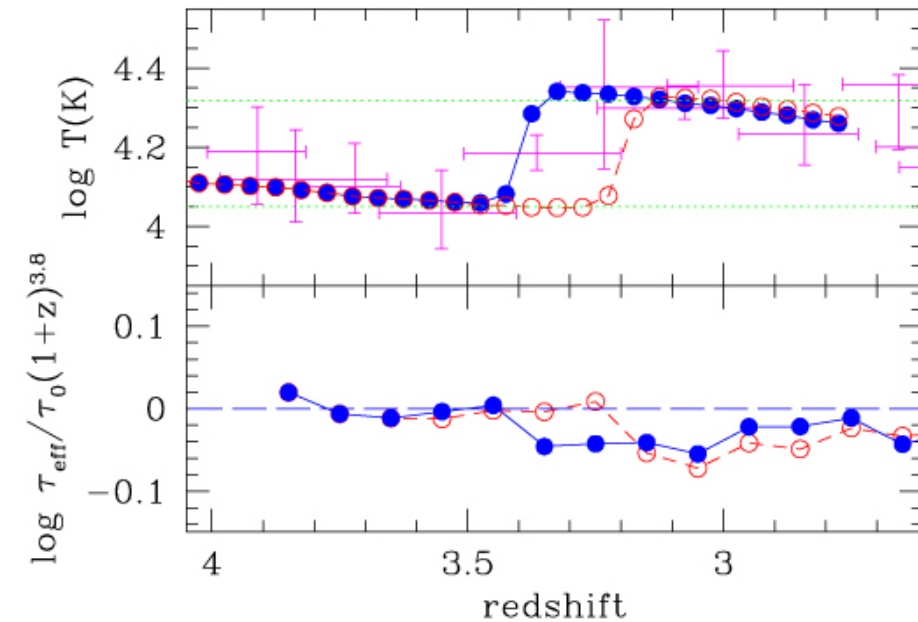
Dall’Aglio *et al.* (2008): 40 UVES spectra (2.6σ only)

But not in these!

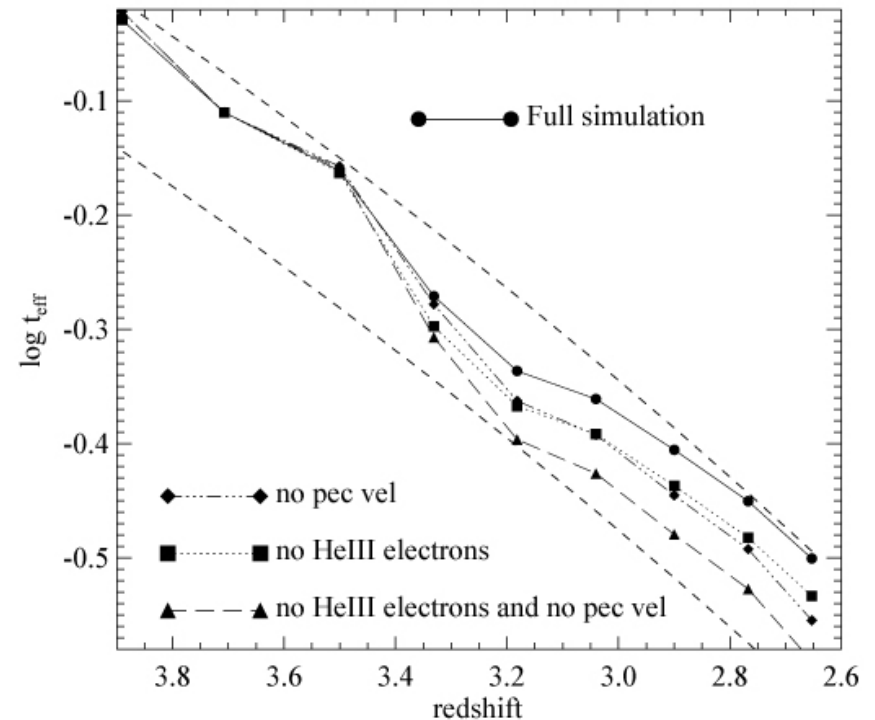
McDonald *et al.* (2005): “Nuisance parameter” in SDSS PS analysis (qualitative only)

Dall’Aglio *et al.* (2009): 1733 SDSS DR5 spectra

Evidence for photo-heating during He II reionisation?



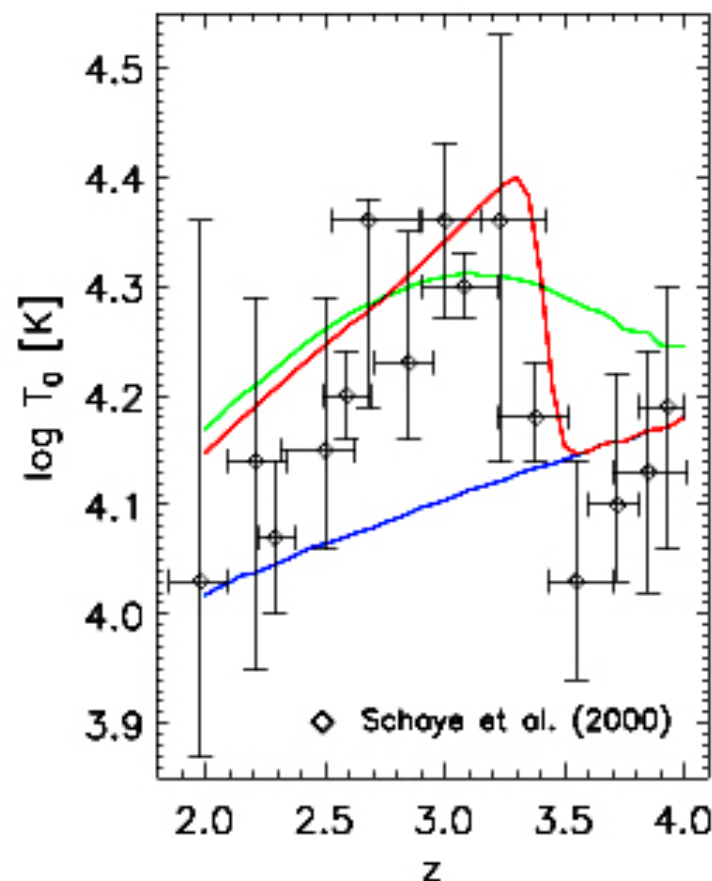
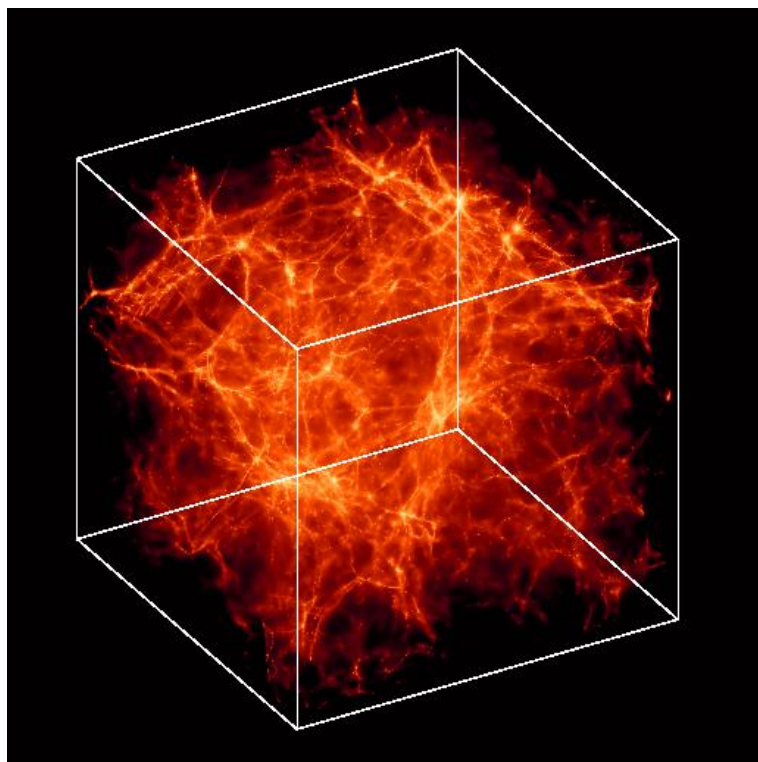
Theuns *et al.* (2002)



Previous explanation from hydro simulations: sudden heating followed rapid recovery due to the hydrodynamical response of the IGM – much quicker than one would expect due to adiabatic cooling following reheating alone.

Simulating the “dip” with GADGET-2

High resolution hydrodynamical simulations, $15h^{-1}$ cMpc, 2×400^3 gas+DM particles, different thermal histories, follow non-equilibrium gas chemistry.



Bolton, Oh & Furlanetto (2009b)

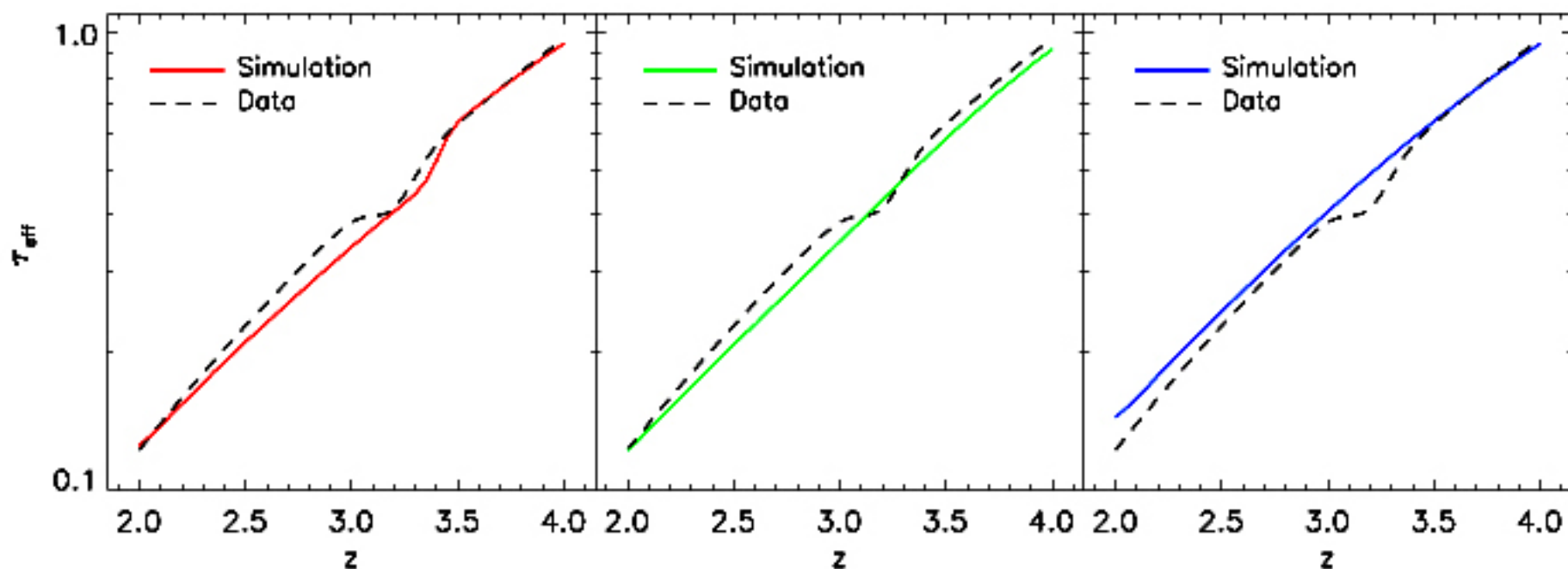
Simulating the “dip” with GADGET-2

Monotonic evolution in all models

$\Delta T \sim 10^4 \text{K}$ at $z=3.4$

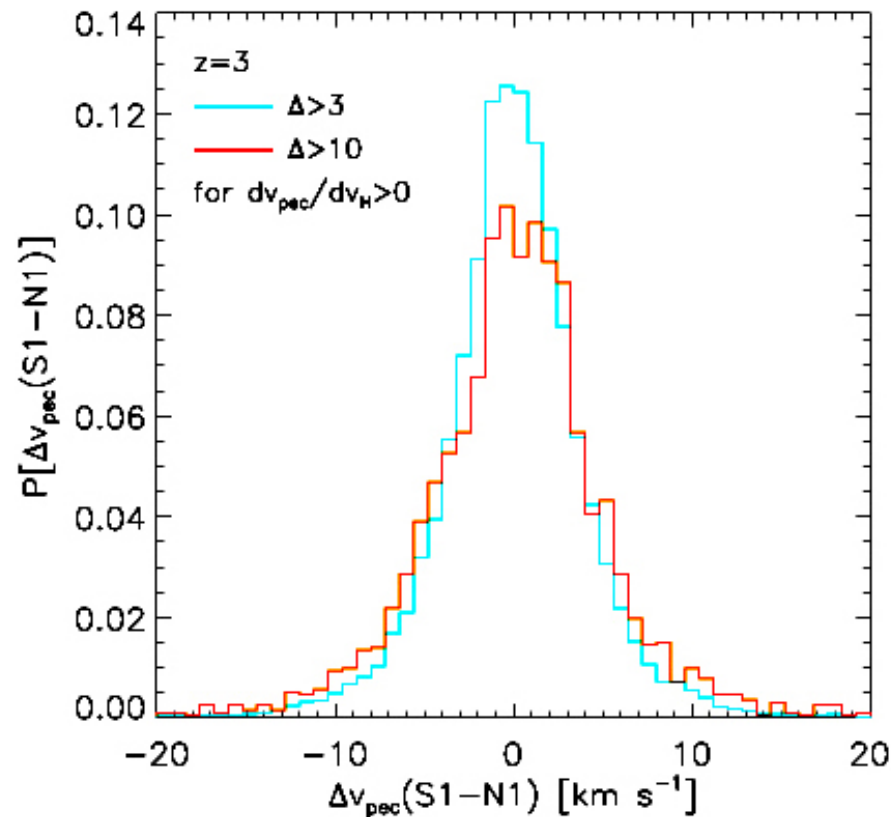
$\Delta T \sim 10^4 \text{K}$ over $\Delta z \sim 2$

No extra heating



Bolton, Oh & Furlanetto (2009b)

Peculiar velocity gradients following reheating



The changes in the peculiar velocities due to expanding gas are small compared to typical line widths in the Ly- α forest (20 km s^{-1}).

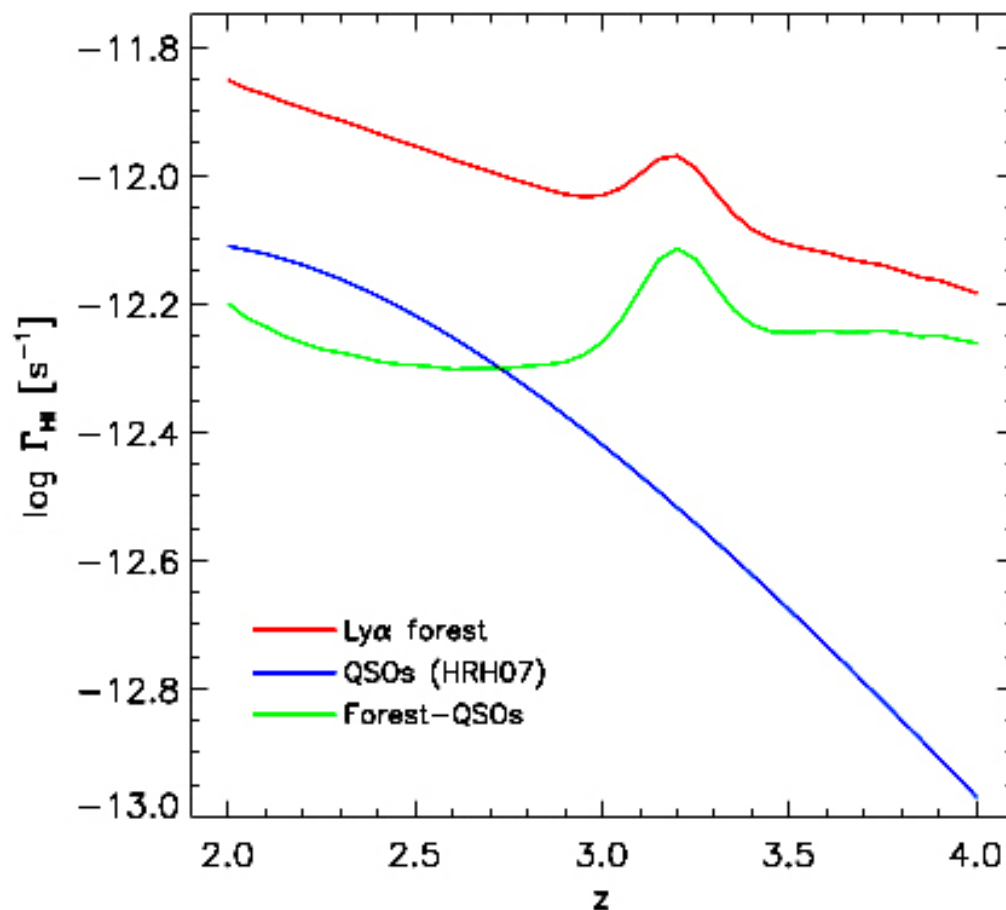
Bolton, Oh & Furlanetto (2009b)

Could it be the HI PI rate?

- The IGM temperature during He II reionisation should increase smoothly over up to several redshift units
- But even with rapid heating, it's hard to reproduce the recovery of the “dip”

What's left?

$$\tau = \tau_0 \frac{(1+z)^6 (\Omega_b h^2)^2}{T^{0.7} H(z) \Gamma(z)} \Delta^2$$



Bolton, Oh & Furlanetto (2009b)

Some (speculative) possibilities...

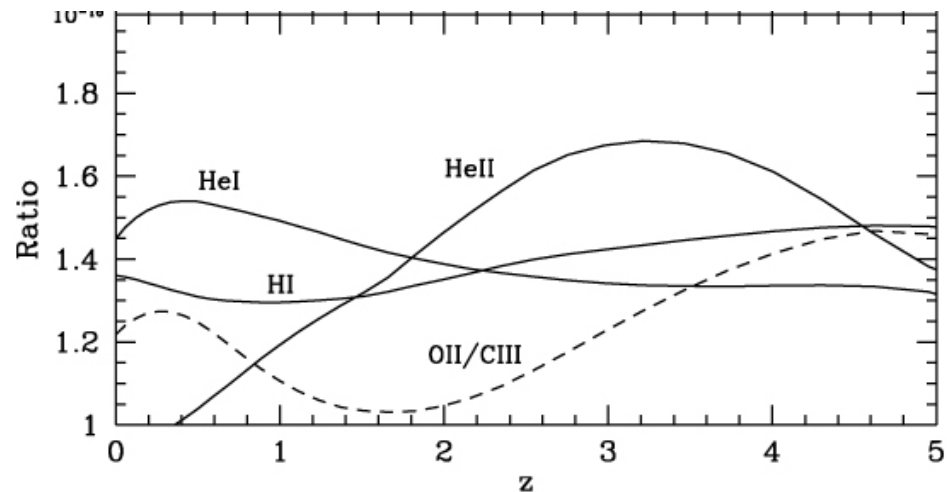
- A modulation in the mean free path for HI ionising photons as HI Lyman limit system expand/contract as the pressure gradient changes following heating. Dynamical time about right, but it requires cosmic variance to be small.

$$\Gamma_{HI} = \int_{\nu_L}^{\infty} \frac{\epsilon_{\nu} \lambda_{\nu} \sigma_{\nu}}{h\nu} d\nu$$

Hot IGM

Cooler LLS

- Recombination radiation - the reprocessing of He II ionising photons into lower frequency radiation could boost the HI photo-ionisation rate towards the tail-end of He II reionisation.



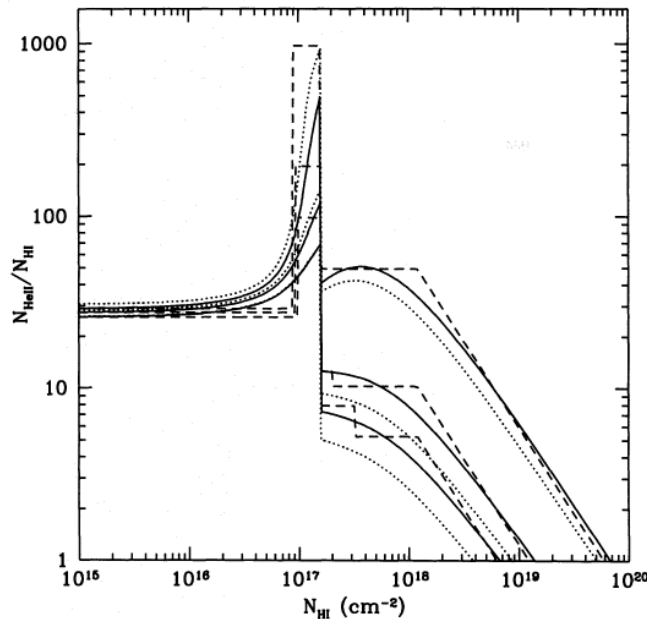
Haardt & Madau (1996),
but see also **Faucher-Giguere et al. (2009)**

Conclusions

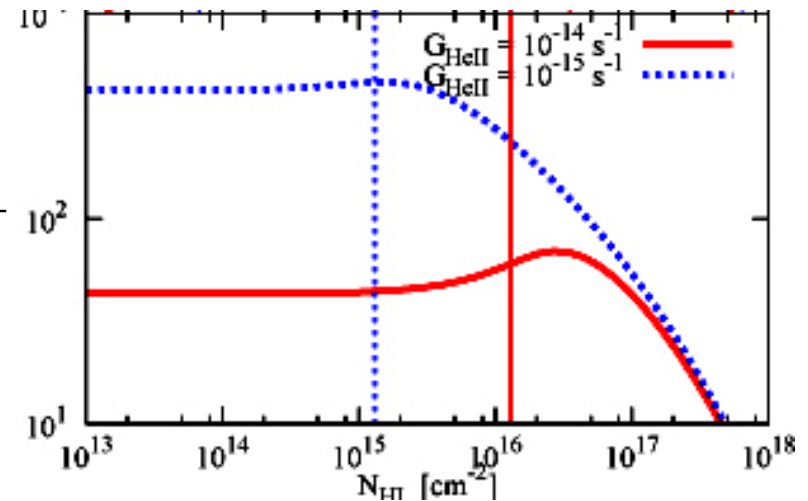
- Photo-heating during He II reionisation is model dependent, but is unlikely to result in large ($\Delta T > 10^4 \text{K}$) **and** rapid ($\Delta z < 0.2$) temperature boosts.
- The dip in the HI effective optical depth cannot be explained by the thermal evolution during He II reionisation – the timescales for heating and cooling are too long.
- The remaining possibility is a modulation of the metagalactic H I photo-ionisation rate.
- Other RT effects associated with He II reionisation could then be responsible – but the explanation (if the feature is real) remains an open question...

He II Lyman limit systems

The He II column density distribution is not observationally constrained. Models must be used to predict the relationship between physical gas density and the observed HI column density distribution.



$$\eta = \frac{N_{\text{HeII}}}{N_{\text{HI}}}$$



McQuinn *et al.* (2009)

Haardt & Madau (1996)

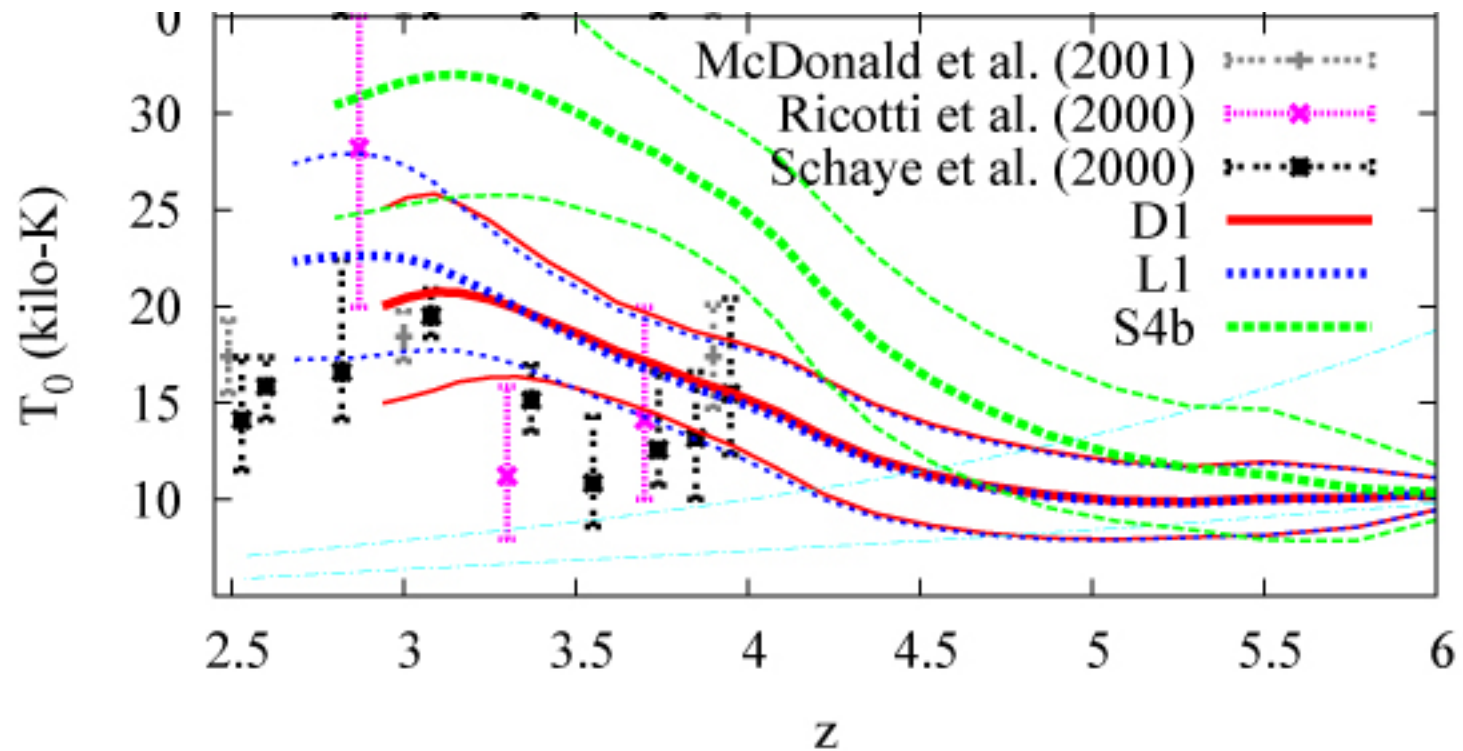
For

$$\lambda \propto \nu^{1.5}, \quad \Delta T \approx 7000 \text{ K}$$

$$\lambda \propto \nu^{2.5}, \quad \Delta T \approx 15000 \text{ K}$$

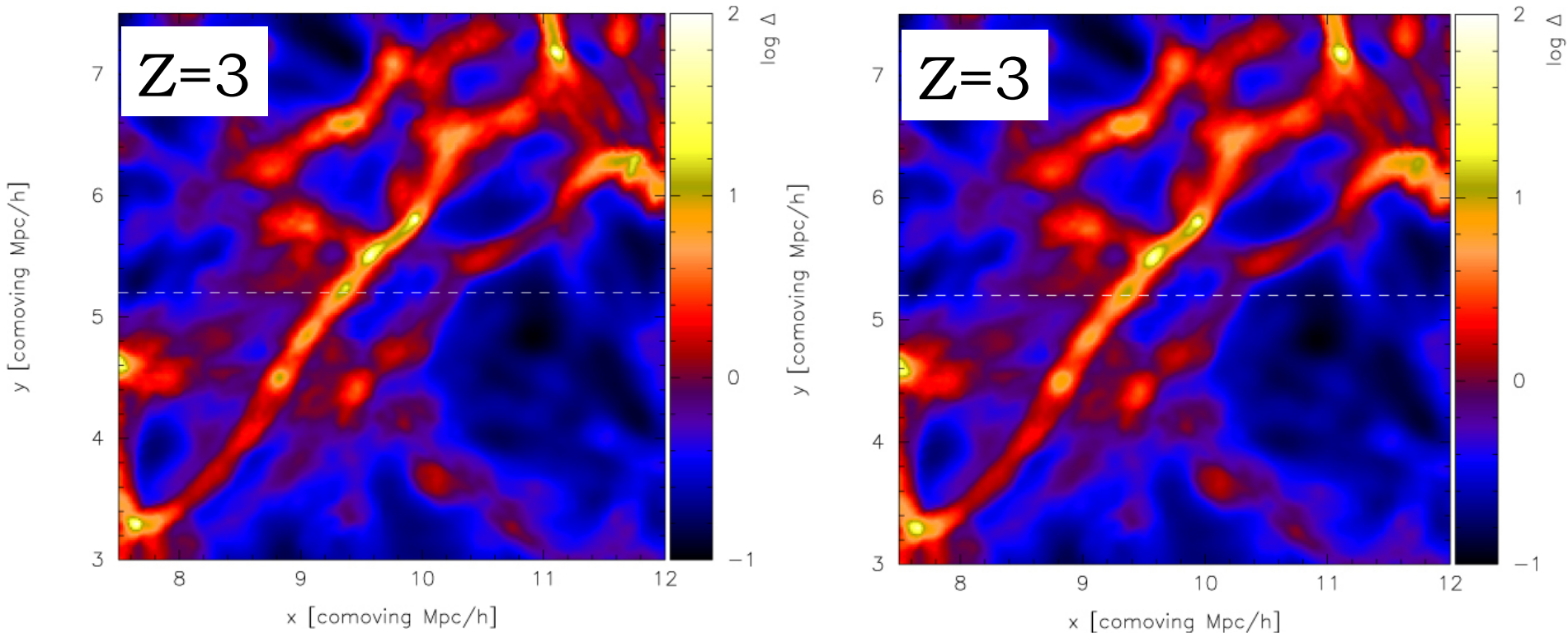
Temperature evolution with redshift

Dark matter + 3D RT in post processing



McQuinn *et al.* (2009)

Gas distribution following reheating



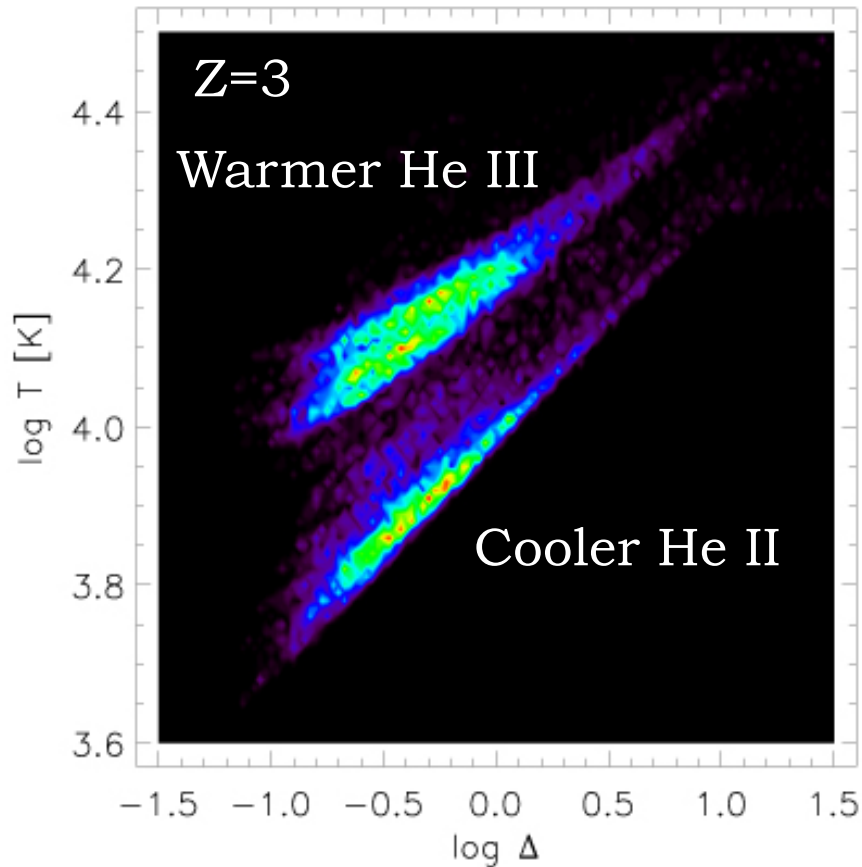
No extra heating

$\Delta T \sim 10^4 \text{ K}$ at $z=3.4$

Pressure smoothing of the density field does not help - $\Delta z \sim 0.2-0.4$ is too short a timescale for the gas to respond in all but the densest regions of the IGM.

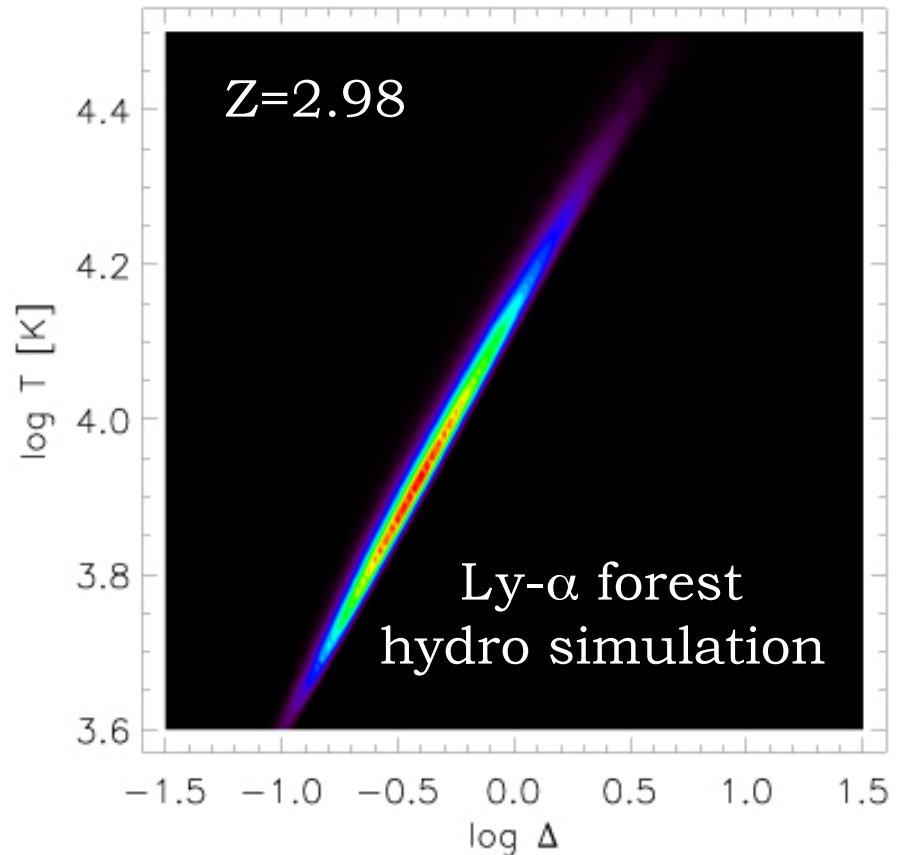
The temperature-density relation

RT: Two-phase, lots of scatter



Bolton, Oh & Furlanetto (2009a)

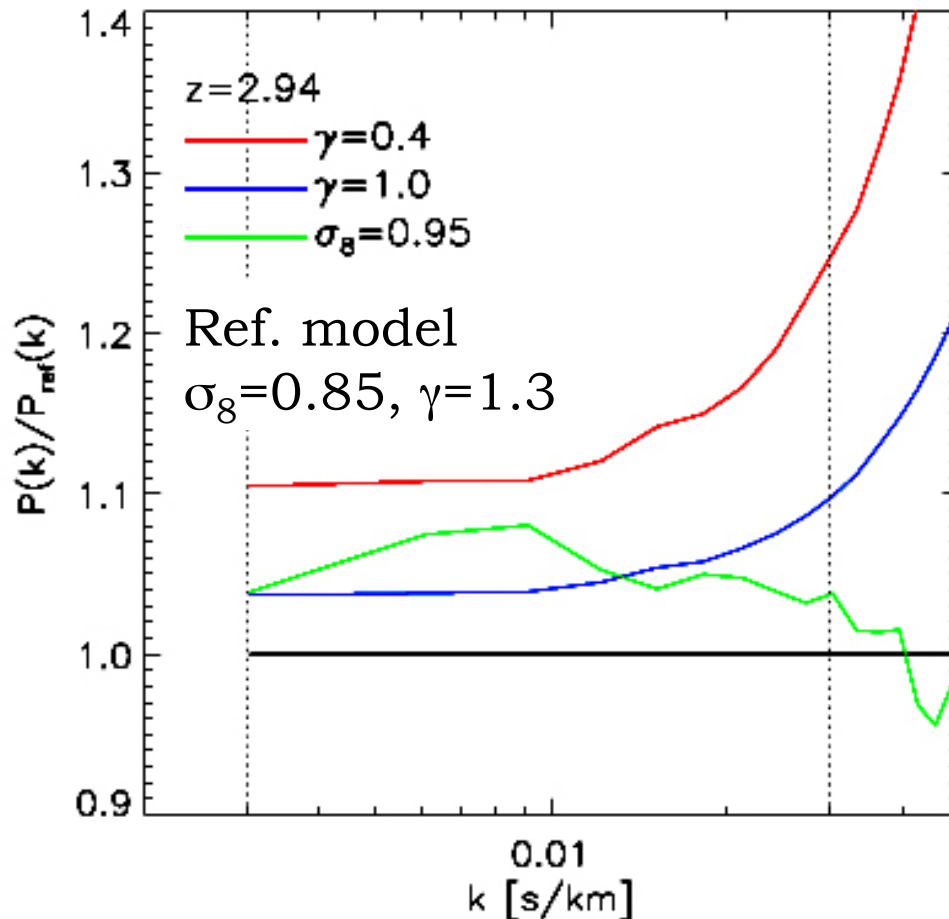
OT: Tight power law $T = T_0 \Delta^{\gamma-1}$



Balance between photo-heating
and adiabatic cooling (**Hui &
Gnedin 1997**)

Possible implications for cosmology with the Ly- α forest?

Ly- α power spectrum



- The Ly- α forest flux distribution favours a mildly inverted T- ρ relation (**Becker *et al.* 2007, Bolton *et al.* 2008**)
- There is a degeneracy between σ_8 and the slope of the T- ρ relation for the flux PS – smaller γ mimics larger σ_8
- Photo-heating and the thermal state of the IGM during He II reionisation will be important to consider when using the Ly- α forest PS to measure the matter power spectrum at small scales.